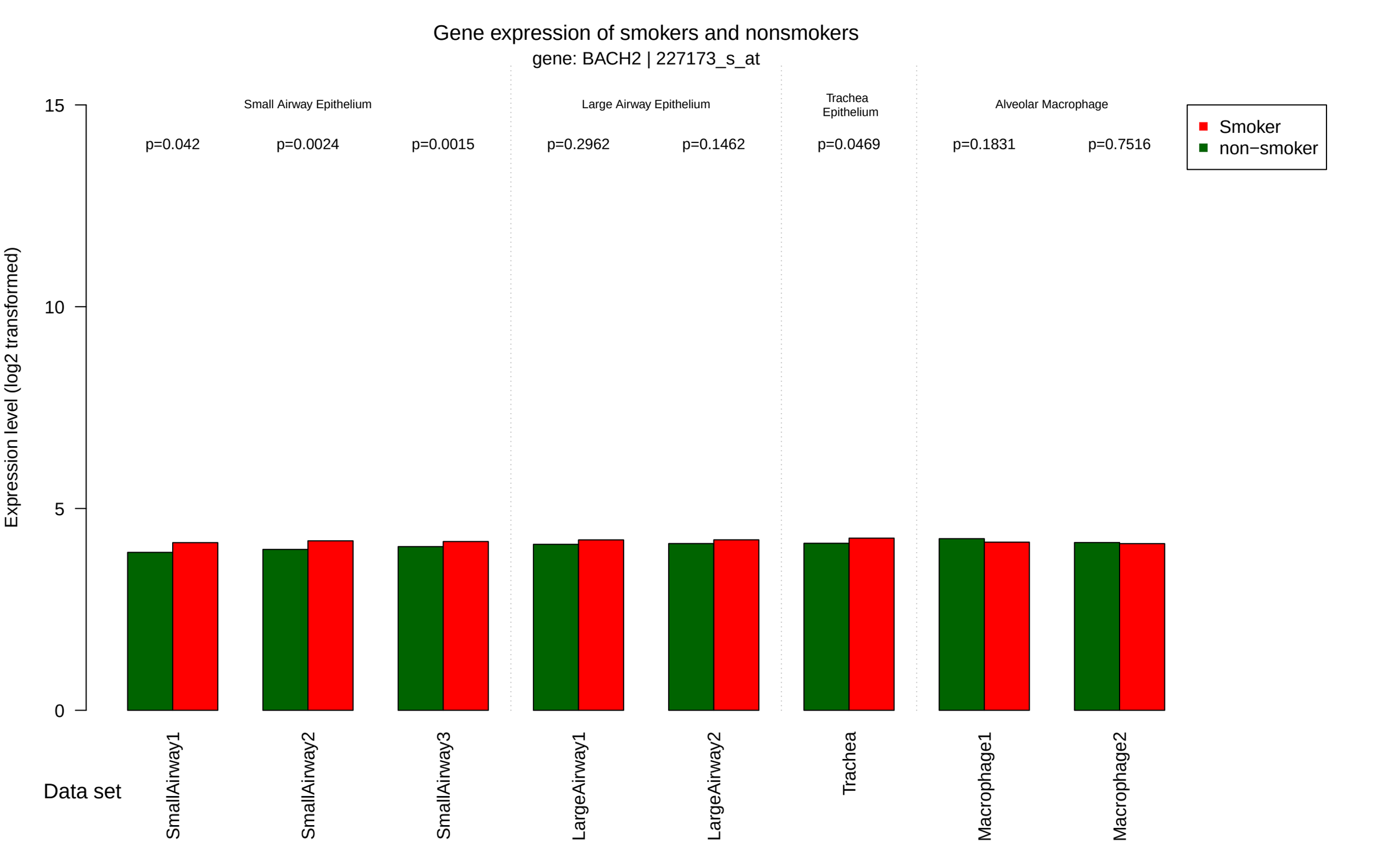
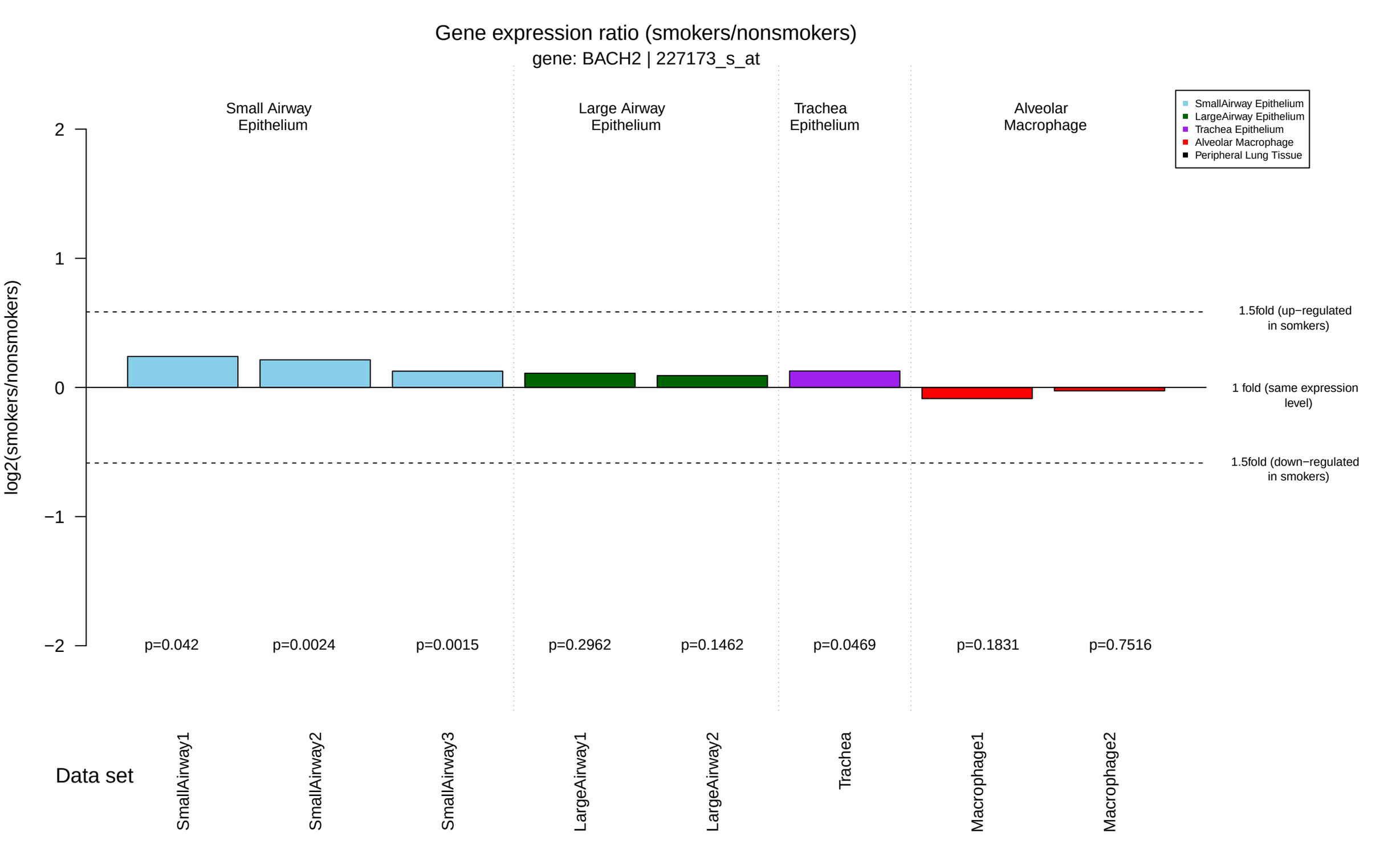
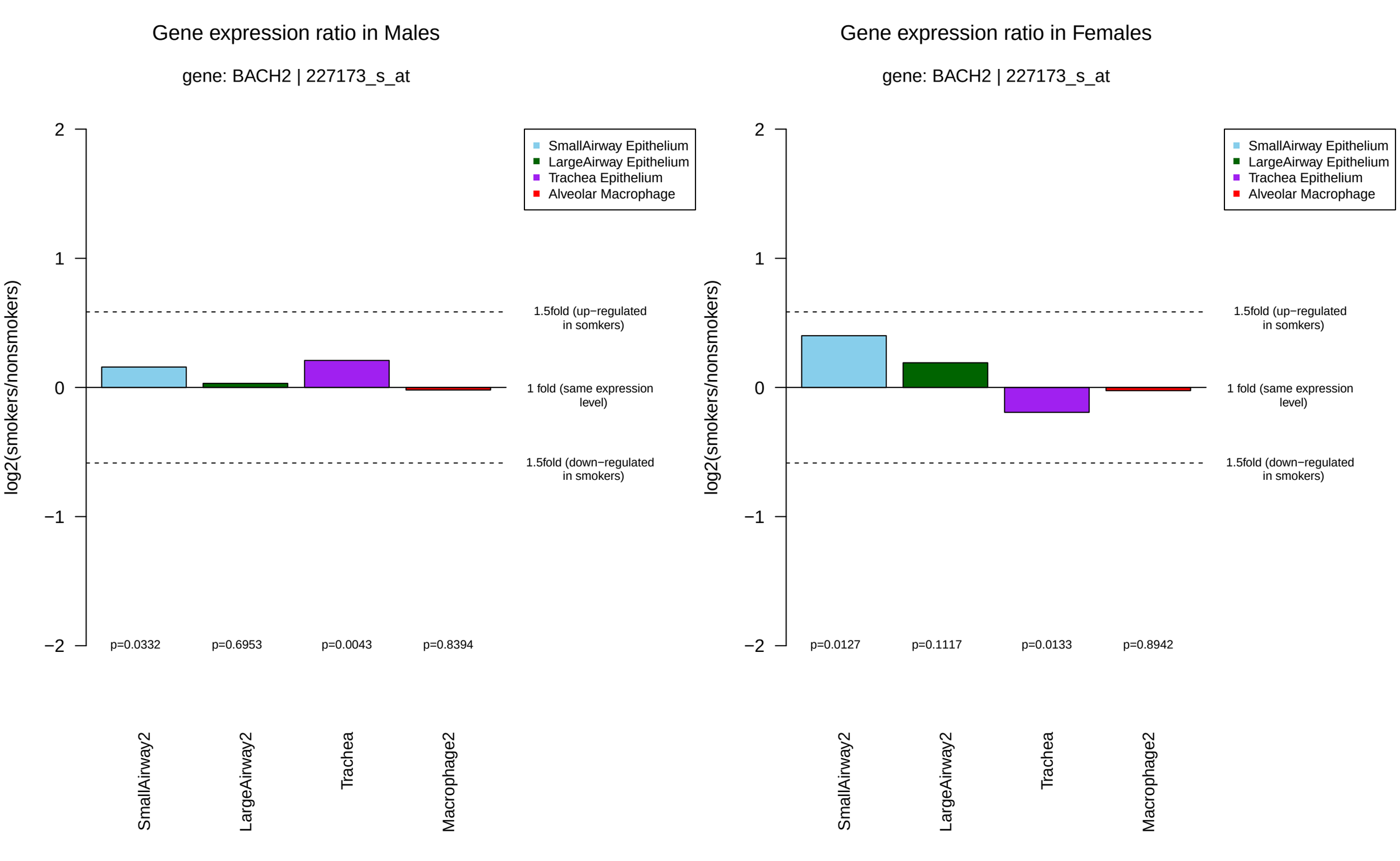
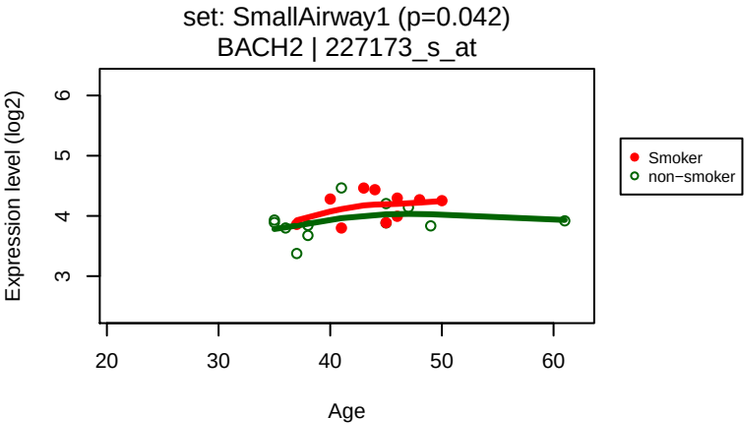
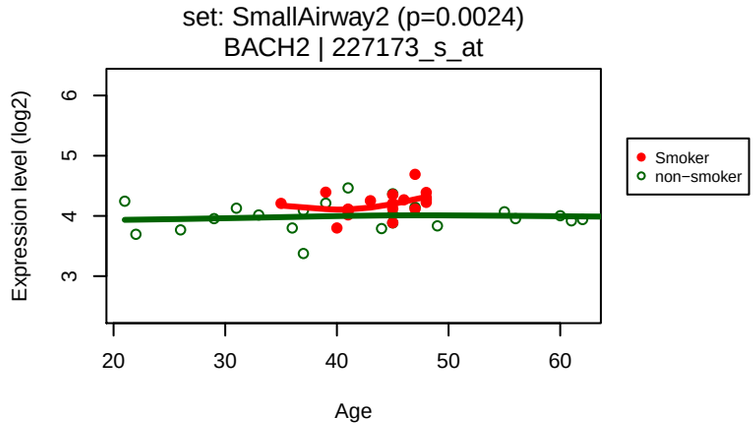
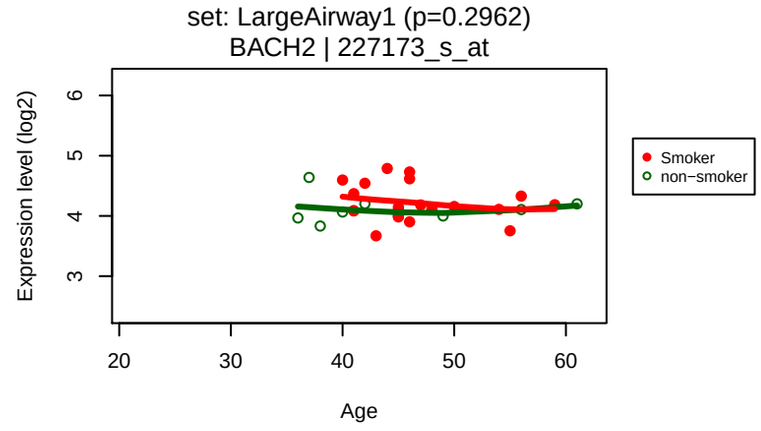

Gene expression of smokers and nonsmokers
gene: BACH2 | 227173_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.042
p=0.0024
p=0.0015
p=0.2962
p=0.1462
p=0.0469
p=0.1831
p=0.7516
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

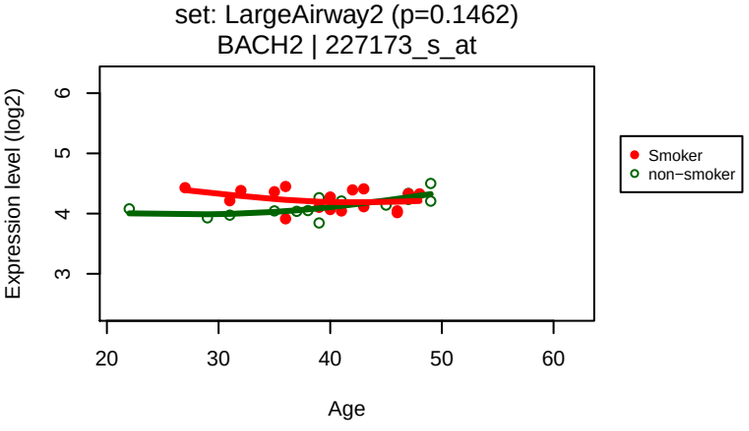

Gene expression ratio (smokers/nonsmokers)
gene: BACH2 | 227173_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.042
p=0.0024
p=0.0015
p=0.2962
p=0.1462
p=0.0469
p=0.1831
p=0.7516
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

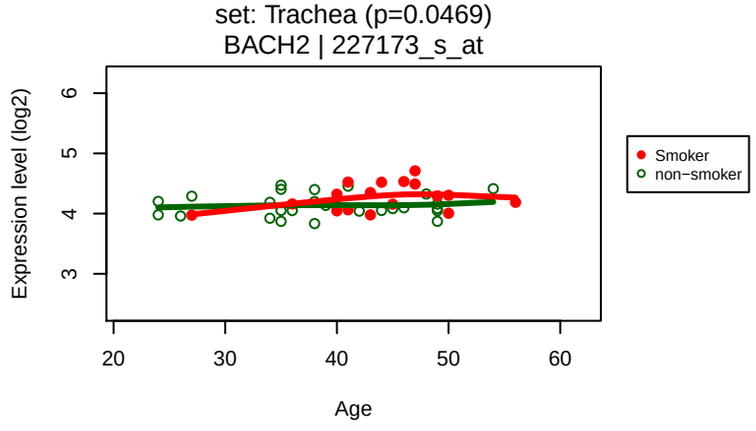

Gene expression ratio in Males
gene: BACH2 | 227173_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0332
p=0.6953
p=0.0043
p=0.8394
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACH2 | 227173_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0127
p=0.1117
p=0.0133
p=0.8942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

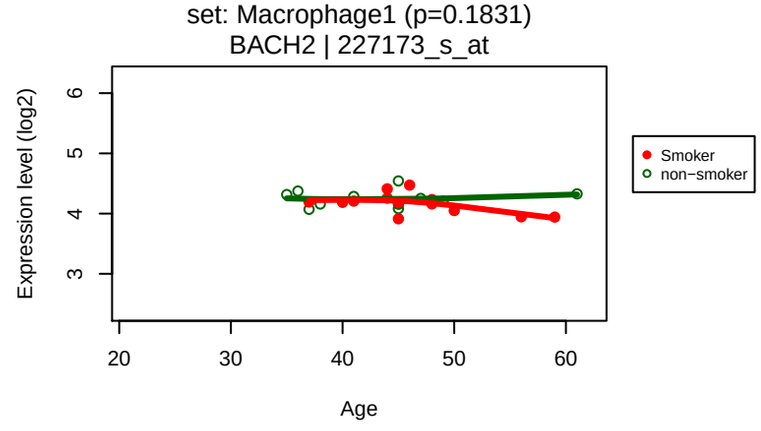

set: SmallAirway1 (p=0.042)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker

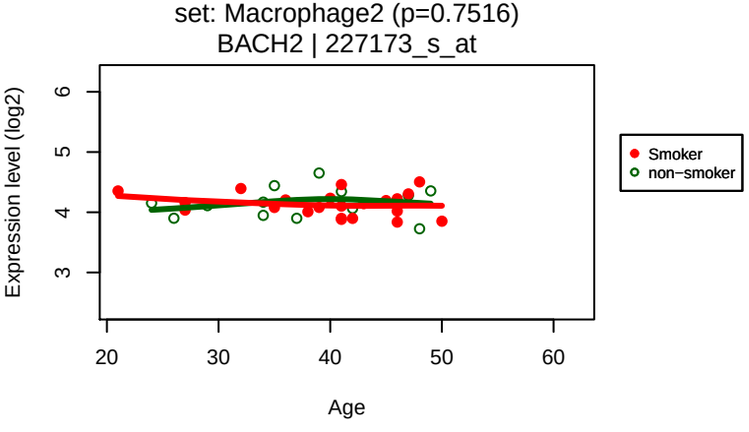

set: SmallAirway2 (p=0.0024)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker

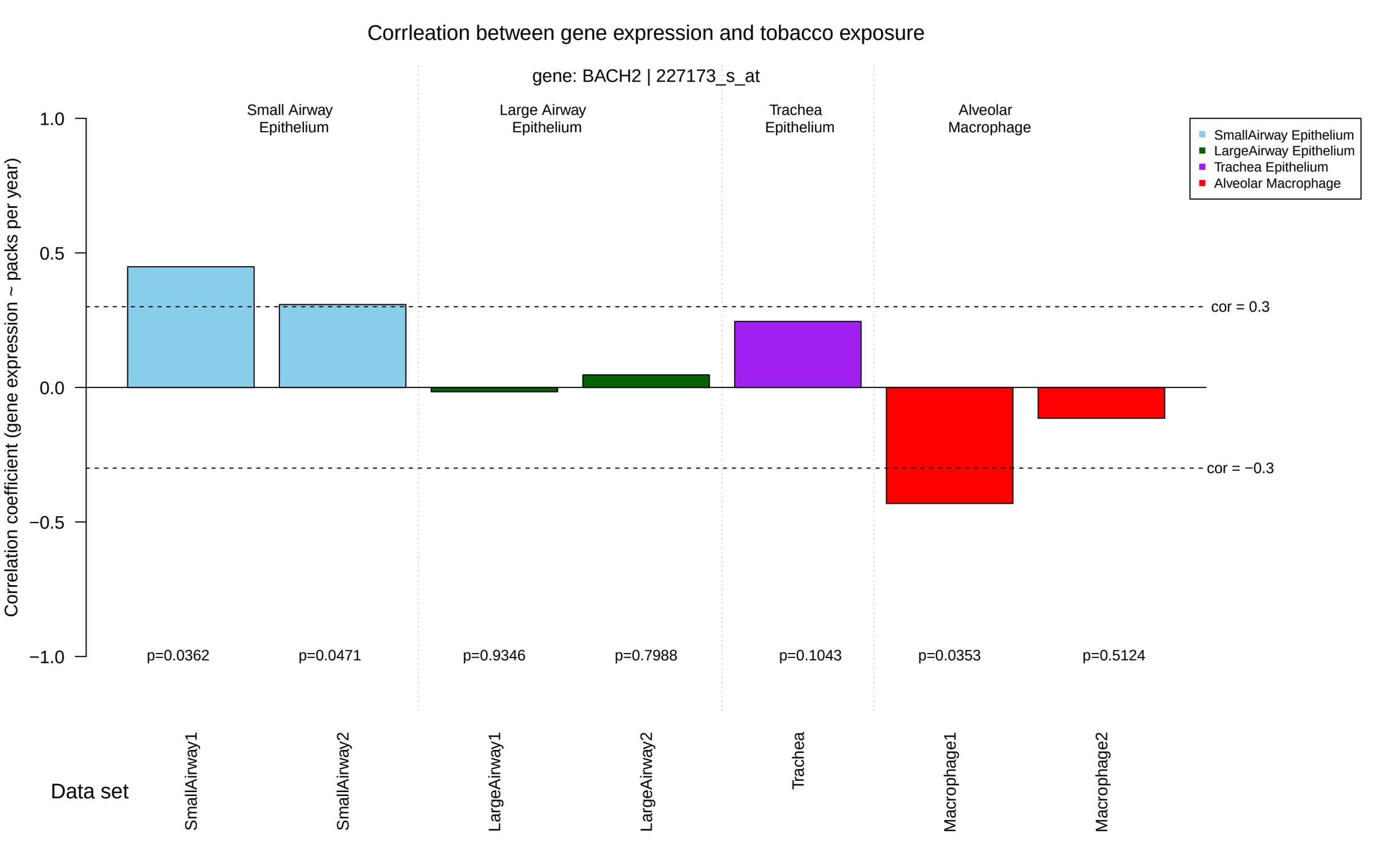

set: LargeAirway1 (p=0.2962)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1462)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0469)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1831)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7516)
BACH2 | 227173_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACH2 | 227173_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0362
p=0.0471
p=0.9346
p=0.7988
p=0.1043
p=0.0353
p=0.5124
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2